

BMJ Open Restoring invisible and abandoned trials of gabapentin for neuropathic pain: a clinical and methodological investigation

Evan Mayo-Wilson ¹, Xiwei Chen,¹ Riaz Qureshi,² Stephanie Dickinson,¹ Lilian Golzarri-Arroyo,¹ Hwanhee Hong,³ Carsten Görg,⁴ Tianjing Li⁵

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ABSTRACT

Introduction Gabapentin (Neurontin) is prescribed widely for conditions for which it has not been approved by regulators, including certain neuropathic pain conditions. There is limited evidence that gabapentin is safe and effective for the treatment of neuropathic pain. Published trial reports, and systematic reviews based on published trial reports, mislead patients and providers because information about gabapentin's harms has been published only partly. We confirmed that trials conducted by the drug developer have been abandoned, and we plan to conduct a restoration with support from the Restoring Invisible and Abandoned Trials Support Centre (<https://restoringtrials.org/>).

Methods and analysis In this study, we will analyse and report the harms that were observed in six trials of gabapentin, which have not been reported publicly (eg, in journal articles). We will use clinical study reports and individual participant data to identify and report the harms observed in each individual trial and to summarise the harms observed across all six trials. We will report all adverse events observed in the included trials by sharing deidentified data and summary tables on the Open Science Framework (<https://osf.io/w8puv/>). Additionally, we will produce a summary report that describes differences between the randomised groups in each trial and across trials for prespecified harms outcomes.

Ethics and dissemination We will use secondary data. This study was determined to be exempt from Institutional Review Board (IRB) review (protocol #1910607198).

INTRODUCTION

Gabapentin (Neurontin) was approved in 1993 by the US Food and Drug Administration (FDA) for the treatment of epilepsy. It was later approved for the treatment of postherpetic neuralgia. To encourage doctors to prescribe gabapentin for many types of neuropathic pain, including 'off-label' indications other than those approved by FDA, the manufacturer published favourable clinical trial results in medical journals.¹ In 2004, the manufacturer pleaded guilty to civil and criminal charges related to illegal

Strengths and limitations of this study

- Gabapentin is prescribed widely and thus of ongoing clinical interest.
- Gabapentin's developer has no plans to publish complete information about its harms, so restoration is needed to complete the published record.
- This study will report previously undisclosed harms in six trials of gabapentin for neuropathic pain.
- Unpublished data to be used in this study will provide a more comprehensive account of the drug's effects compared with previous journal articles and systematic reviews limited to published evidence.

marketing and paid US\$430 million to the US Department of Justice.² Pfizer, the manufacturer that ultimately acquired the drug, paid US\$325 million in 2014 to settle plaintiffs claims that it defrauded patients and benefit providers in this manner.³

Gabapentin continues to be prescribed widely,⁴ including for the treatment of pain, perhaps because published trial reports and systematic reviews based on published trial reports continue to mislead patients and providers about its benefits and harms. In published journal articles, undisclosed changes to primary outcomes and methods of analysis contributed to overestimating gabapentin's potential benefits.⁵ Such changes were possible because the manufacturer conducted multiple analyses of primary and secondary outcomes and reported only a subset of the results.^{6 7} Publications also include very little information about harms ('adverse events' (AEs)) that occurred in clinical trials,^{8–10} which patients with chronic pain say are critically important to their decisions about whether to take drugs to treat pain.¹¹ More information about benefits and harms can be found in unpublished clinical study reports (CSRs), which the manufacturer



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For numbered affiliations see end of article.

Correspondence to

Dr Evan Mayo-Wilson;
emayowil@iu.edu

Table 1 Glossary of terms related to adverse events and sources

Term	Definition used in our study
Clinical study report (CSR)	A special document type originating with drug and device manufacturers for submitting information to regulators (eg, US Food and Drug Administration (FDA), European Medicines Agency). CSRs usually follow intentional guidelines for structure and content, and they often contain detailed summaries of trial design (including the trial protocol and statistical analysis plan) and results. ⁸²
Individual patient data (IPD)	A record of variables collected for each participant in a clinical trial (eg, clinical characteristics, scores on tests and questionnaires), usually stored in a database file.
Adverse event (AE)	The International Conference on Harmonisation defines an 'AE' as 'any untoward medical occurrence in a patient or clinical investigation subject administered a pharmaceutical product and which does not necessarily have to have a causal relationship with this treatment.' ⁸³ The US FDA and other regulators use this definition. ^{84 85}
Coding Symbols for a Thesaurus of Adverse Reaction Terms (COSTART)	The Coding Symbols for a Thesaurus of Adverse Reaction Terms (COSTART) is a terminology developed and used by the FDA for coding, filing and retrieving of post-marketing adverse drug/biologic experience reports. ³³ COSTART was replaced by Medical Dictionary for Regulatory Activities in the late 1990s. ³⁴
Preferred term (COSTART symbol)	A 20-character code used to identify events using the COSTART system. ³³
Mid-level systems	According to the COSTART manual, a mid-level system is a 'mid-level pathophysiologic classification of COSTART for purposes of categorising and retrieving information based on disease associations.' 'This section is hierarchical in arrangement, allowing one to be very general or more specific and is a convenient strategy for searching for drug-induced disease'. ³³
Body systems	According to the COSTART manual, 'Essentially anatomic, this body system classification is sometimes the basis of search strategy. The classification is hierarchical in nature.' ³³
Serious	The US FDA and other regulators consider AEs as 'serious' when they lead to or prolong hospitalisation, cause death or disrupt normal life functions. ^{84 85} 'Serious' is not synonymous with 'severe,' the latter being a descriptive characteristic rather than a regulatory classification.
Time under observation	The length of time (eg, days) during which a participant could have reported harms to study investigators, which we estimated for this study using multiple data sources.

Definitions adapted from previous reports of the MUDS study.^{7-10 16}
MUDS, Multiple Data Sources for Meta-Analysis.

called 'Research Reports', and individual patient data (IPD) that became available during litigation (table 1). Despite the availability of these data, information about harms has not been incorporated in reviews and guidelines about gabapentin, such as a highly cited Cochrane review,^{12 13} that continue to be used to promote the use of gabapentin.¹⁴

We confirmed that trials conducted by the manufacturer have been abandoned.¹⁵ We contacted Pfizer in 2015 to share the published protocol for the Multiple Data Sources for Meta-Analysis (MUDS) study, a methodological study about the use of different data sources for systematic reviews and meta-analysis.¹⁶ We subsequently exchanged emails and spoke by telephone with Pfizer representatives. In 2016, Pfizer confirmed by telephone that it has no plans to publish additional data or analyses from trials of gabapentin for neuropathic pain. In 2017, we received an unexpected email from a Pfizer contractor in response to our earlier request for information about gabapentin trials; we replied, but Pfizer did not respond (online supplemental file 1). Thus, we issued a 'call to action' to produce a complete account of the harms that

were observed but not reported in six important clinical trials,¹⁷ for which we later received support from the Restoring Invisible and Abandoned Trials Support Centre.¹⁸

Objectives

Our objective in this study is to report the harms that were observed in six trials of gabapentin, which have not been reported publicly (eg, in journal articles). We will use CSRs and IPD to identify and report the harms observed in each individual trial and to summarise the harms observed across all six trials.

This study was determined to be exempt from IRB review (IRB Protocol #: 1910607198).

METHODS AND ANALYSIS

Patient and public involvement

This study will use data that were previously collected by the developer and manufacturer of gabapentin. Members of our team contributed to data harmonisation as part of the MUDS study; patients were involved in the design,

conduct, reporting and dissemination of the MUDS study. Patients and the public will not be involved in this additional investigation.

Outcomes

We will report all AEs observed in the included trials by sharing deidentified data and summary tables on the Open Science Framework (<https://osf.io/w8puv/>).

Additionally, we will attempt to produce a summary report that describes differences between the randomised groups in each trial and across trials for the following outcomes:

1. Proportion of participants who experienced one or more AEs.
2. Proportion of participants who experienced one or more serious AEs.
3. Rate of one or more AEs.
4. Rate of one or more serious AEs.
5. Proportion of participants who discontinued their assigned intervention because of AEs.
6. Proportion of participants who discontinued their assigned intervention for any reason.
7. Time to discontinuation because of AEs.
8. Time to discontinuation for any reason.
9. Proportion of participants who experienced specific AEs at the level of preferred term, mid-level system, and body system.
10. Rate of specific AEs at the level of preferred term, mid-level system and body system.
11. Time to specific AEs at the level of preferred term, mid-level system and body system.

Eligible trials

This is an additional study using data from the MUDS study, which examined two drugs, gabapentin and

quetiapine. The MUDS investigators selected drugs for which they had access to both public and non-public sources of information; they searched for public and non-public information, and they requested additional information from the manufacturers, as described previously.^{7 16 19}

This study focuses on gabapentin only. The MUDS study included 21 parallel randomised clinical trials comparing gabapentin with placebo for neuropathic pain in adults. Crossover studies were excluded. Six trials included in the MUDS study were conducted by the manufacturer and are included in this study (table 2); for these trials, we have access to CSRs and IPD^{20–27} as well as public data sources, such as journal articles.^{28–31} Pfizer-released information about these studies during litigation in which Kay Dickersin, principal investigator of the MUDS study, served as an expert witness. During litigation, Pfizer provided to plaintiff's attorneys: a list of trials of gabapentin, internal company documents (Inferential Analysis Plans, Research Reports and memos), and Microsoft Access databases containing IPD. In response to requests for additional information and meta-data (eg, codebooks), Pfizer confirmed that materials not released during litigation either never existed or were lost (online supplemental file 1).

Data collection and management

Obtaining aggregate data

From each report of each eligible trial in the MUDS study, two investigators independently extracted data using the open access Systematic Review Data Repository (<http://srdhr.ahrq.gov/>) and resolved discrepancies by consensus and through discussion with a third reviewer if necessary. The MUDS investigators shared the statistical code and

Table 2 Eligible placebo controlled trials

Study protocol number	Pain condition(s) included	No of participants assigned to gabapentin and placebo	Daily dose in each gabapentin group
945-210 ²⁰	Diabetic neuropathic pain	165	Maximum (target) dose 3600 mg
945-224 ²¹	Diabetic neuropathic pain	325	Fixed doses 600, 1200 and 2400 mg/day
945-400-211 ^{22–24}	Postherpetic neuralgia	229	Fixed dose 3600 mg/day
945-430-295 ²⁵	Postherpetic neuralgia	334	Fixed doses 1800 and 2400 mg/day
945-430-306 ²⁶	Neuropathic pain (Allowed: complex regional pain syndrome or reflex sympathetic dystrophy; pain because of traumatic injury; diabetic peripheral neuropathy; phantom limb pain or pain following amputation of limbs; post-herpetic neuralgia; radicular pain or radiculopathy associated with spinal stenosis; stroke)	305	Maximum (target) dose 2400 mg
A945-1008 ²⁷	Diabetic peripheral neuropathy	389	Maximum (target) dose 3600 mg

Table 3 Variables for each participant in the MUDS database

Variable name	Description
study_id	Study identification number as assigned by the manufacturer.
patient_id	Participant identification number as assigned by the manufacturer.
sex	Participant sex, harmonised to 'female' or 'male' by the MUDS investigators.
treatment	Treatment group allocation, including dose, as reported by the manufacturer.
treatment_dic	Treatment group allocation, recoded by the MUDS team as either 'placebo' or 'gabapentin'.
pain_d01 pain_d02 ... pain_d98	Daily pain score. Each morning on arising, participants wrote down a number to rate their pain during the previous 24 hours on an 11-point Likert scale ranging from 0 (no pain) to 10 (worst possible pain).
sleep_d01 sleep_d02 ... sleep_d98	Daily sleep score. In five of six trials, ^{20–25} each morning on arising, participants wrote down a number to rate how pain interfered with sleep on an 11-point Likert scale. The scale ranged from 0 (did not interfere with sleep) to 10 (completely interferes with sleep). The daily sleep score was not collected in one trial. ²⁶

MUDS, Multiple Data Sources for Meta-Analysis.

datasets from the MUDS study on the Dryad repository.³² The MUDS investigators shared partially redacted CSRs on the Drug Industry Documents Archive,^{20–26} which are complete except for appendices containing identifying information (eg, patient initials, date of birth, exact dates of medical examinations), which the MUDS investigators did not have resources to redact and recode.

Obtaining IPD

The MUDS study used CSRs and partially deidentified IPD that were provided as Microsoft Access Databases without codebooks to Professor Kay Dickersin for her expert witness report in litigation against Pfizer. The MUDS investigators developed codebooks and harmonised the databases by comparing the databases with case report forms (which show how and when data were recorded) and statistical analysis plans (which show how data were coded and analysed) to identify the variables contained in the databases.

In this study, we will use a subset of the IPD database that was harmonised by the MUDS investigators. The MUDS database includes two types of tables, which include information about participants and AEs, respectively (tables 3 and 4).

Individual harms and groups of harms

Multiple systems may be used to classify and analyse harms including Coding Symbols for a Thesaurus of Adverse Reaction Terms (COSTART), Medical Dictionary for Regulatory Activities (MedDRA), Systemized Nomenclature of Medicine (SNOMED), or Common Terminology Criteria for Adverse Events (CTCAE).^{33–36} Such systems

Table 4 Variables for each AE in the MUDS database

Variable name	Description
COSTARTsymbol	For five of six trials, ^{20–26} COSTART symbol ('preferred term') for each AE, which the MUDS team matched to alphanumeric codes reported by the manufacturer. One trial ²⁷ was not coded using COSTART, and we will match AE terms to COSTART symbols for this study.
COSTARTmid1	For five of six trials, ^{20–26} COSTART mid-level system for each AE as matched by the MUDS team. One trial ²⁷ was not coded using COSTART, and we will match AE terms to COSTART symbols for this study.
COSTARTbodyA1	For five of six trials, ^{20–26} COSTART primary body system for AE as matched by MUDS team. One trial ²⁷ was not coded using COSTART, and we will match AE terms to COSTART symbols for this study.
aetext	Text describing adverse events as reported by the manufacturer.
aestartday	Time of onset as reported by the manufacturer (days from start of medication to start of AE).
aeendday	Time of resolution as reported by the manufacturer (days from start of medication to end of AE).
recurrent	For five of six trials, ^{20–26} whether the AE recurred as reported by the manufacturer. Recurrence was not available in one of the six trial databases. ²⁷
severity	Severity of the AE as reported by the manufacturer (mild, moderate, severe).
serious	Whether AE was considered serious as reported by the manufacturer.
releat_dic	Whether the AE was judged as related to treatment or caused by treatment. Harmonised by the MUDS investigators as either 'yes'; 'no' or 'insufficient Information'.
action	Action taken following AE, as reported by the manufacturer ("None"; "Dose reduced"; "Dose interrupted"; "Discontinued"; or "Increased").

AE, adverse event; COSTART, Coding Symbols for a Thesaurus of Adverse Reaction Terms; MUDS, Multiple Data Sources for Meta-Analysis.

generally use a hierarchical structure with higher-order terms denoting the anatomic or physiological systems that are affected. Lower order terms typically denote the specific harms experienced. Because some harms are rare, grouping harms by anatomic or physiological systems can increase statistical power and increase the possibility of detecting drug-induced harms. Grouping can also disguise important harms by combining them with less important harms (eg, 'migraine' might be more severe than 'headache', but the distinction would be lost if combined under the higher order term 'headaches').

The COSTART was developed by FDA, and it was being used for regulatory trials when gabapentin was developed.³³ MedDRA replaced COSTART in the late 1990s.³⁴

Using the COSTART system, analyses may be performed at the level of 'preferred term,' which is the lowest level for analysis in the hierarchy, and preferred terms may be grouped for analysis using the following higher levels of aggregation:

- ▶ Body systems.
- ▶ Body system subcategories.
- ▶ Mid-level system.
- ▶ Mid-level system subcategories.

In this study, we plan to assess harms at the level of preferred terms and grouped to mid-level and body systems. For preferred terms that could be matched to more than one body system, we will use the primary body system as recorded by the MUDS investigators.

Mapping individual patient data to the COSTART system

For five included trials, the IPD received from Pfizer included a description of each AE alongside a 5-character alphanumeric 'COSTART code'. After deduplication, the MUDS investigators identified 246 unique 5-character alphanumeric codes that were matched to COSTART preferred terms for analysis using the following methods:

1. In addition to the five-character alphanumeric code, the IPD for one trial²²⁻²⁴ contained a field with the COSTART preferred term. The MUDS investigators used the IPD to match 120 (49%) of the 246 unique five-character alphanumeric codes to preferred terms.
2. IPD for two gabapentin trials^{37 38} that were not eligible for the MUDS study contained both five-digit alphanumeric codes and COSTART preferred terms. The MUDS investigators used these IPD to validate the previously identified preferred terms and to match 33 (10%) additional alphanumeric codes to COSTART preferred terms.
3. An appendix in a CSR³⁹ included both five-digit alphanumeric codes and COSTART preferred terms. Using the software ABBYY FineReader,⁴⁰ the MUDS investigators extracted these data into a spreadsheet, which they used to match 19 (8%) additional five-character alphanumeric codes to COSTART preferred terms.
4. The MUDS investigators converted Index D of the COSTART Dictionary (Glossary of Included Terms) into a spreadsheet to match the remaining five-digit alphanumeric codes to COSTART preferred terms. This was ac-

complished for each five-character alphanumeric code by comparing the accompanying 'AE text' (ostensibly, what was written on the Case Report Form (CRF) to describe each AE) to the terms listed in the COSTART Glossary.³³ The COSTART Glossary includes approximately 6000 synonyms for COSTART preferred terms. This information was used to match 24 (10%) additional five-character alphanumeric codes to COSTART preferred terms.

5. Then the MUDS investigators matched 5 (2%) five-character alphanumeric codes and accompanying text to MedDRA, a more current AE coding system, and mapped them to the corresponding COSTART terms.
6. Finally, three clinicians worked in pairs to review the remaining 45 five-character alphanumeric codes, along with accompanying text and the COSTART Glossary,³³ and to propose appropriate preferred terms. Each of the three clinicians was given 30 five-character alphanumeric codes; the pairs compared their ratings and sought input from the third clinician to resolve any disagreements. The MUDS investigators successfully matched 34 (14%) five-character alphanumeric codes to preferred terms in this manner.
7. The remaining 11 (5%) alphanumeric codes could not be matched.

In one trial,²⁷ some AEs were recorded using terms that mapped to COSTART preferred terms while other terms did not map to COSTART. To analyse this trial and to compare it with the other trials in our study, we will attempt to map all terms to COSTART.

The first step in mapping these terms was performed automatically by the MUDS investigators in Stata (V.13) using the COSTART dictionary and the other included trials. This yielded a match for some entry terms, but many still require manual mapping. To manually map terms, two independent investigators will compare the AE text to COSTART preferred terms. Discrepancies will be reviewed by a third investigator and discussed, and a clinician will review any remaining unmapped terms.

To manually map terms, each of two independent investigators will use the BioPortal website (fifth Edition COSTART)⁴¹ to compare AE text to COSTART preferred terms. BioPortal includes all entry terms and synonyms for preferred terms and also provides an intuitive presentation of the mappings to higher order terms. When mapping the AE text to preferred terms, we will extract the corresponding preferred term ('notation') and the 'prefLabel' (figure 1).

For all mapped preferred terms in trial A945-1008²⁷ that also appear in one of the other five trials, the previously mapped mid-level system and body system will be used in our study. For all preferred terms that are unique to A945-1008,²⁷ we will assign the mid-level system and body system terms that are more specific to the preferred term.

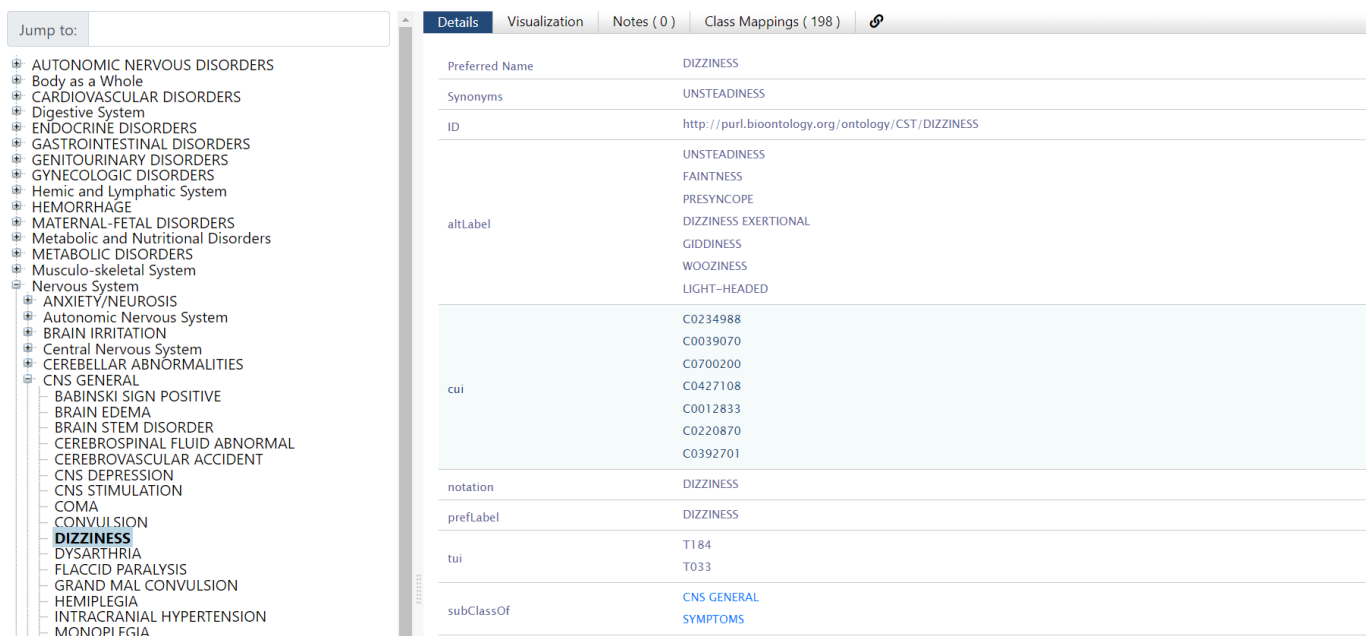


Figure 1 BioPortal COSTART dictionary result for 'Dizziness'. The BioPortal dictionary can be navigated manually on the left-hand side of the figure, or by searching for specific terms using the search function. To the right, the preferred term ('notation'), label ('prefLabel') and alternative labels ('altLabel') appear. The notation, prefLabel and altLabel are 'entry terms' that direct to the preferred term; for example, a physician could write 'light-headed' in their notes, which would be mapped to the preferred term 'Dizziness'. On BioPortal, entering an altLabel into the search bar will bring up the corresponding preferred term. The results on the right also include corresponding mid-level systems and body systems ('subClassOf'). Here, 'Dizziness' maps to 'CNS General' (mid-level) which falls under 'nervous system' (body system), as can be seen by looking on the left side of the page or by clicking on the link to CNS General (in blue). 'Dizziness' also maps to 'symptoms' (mid-level) which falls under 'non-specific disorders' (body system). COSTART, Coding Symbols for a Thesaurus of Adverse Reaction Terms

Methods of analysis in each individual trial

Time under observation for each participant

IPD received from the manufacturer did not appear to include a variable indicating the time during which each participant was under the observation. We will calculate time under the observation for each participant by checking the following sources of information. Where more than one value is available, we will choose the longest valid time (table 5).

For example, if a participant's pain and sleep records were available up to day 46 (then missing), the last recorded medication was taken on day 48, and the participant reported AEs on days 9, 42 and 62, we would estimate the time under observation as 62 days. We will check for potentially invalid values (eg, 620 days) before performing the analysis.

Study discontinuation (drop-out) status for each participant

IPD received as databases from the manufacturer did not include a variable indicating whether each participant had completed the study or discontinued. To compare dropout between groups, we will use information from the CSRs or calculated using the methods described below.

- ▶ One CSR²¹ listed participant identifiers for participants who discontinued.
- ▶ Four CSRs^{20 22–26} listed each participant's end-of-study status as 'Completed study/phase' or giving a reason

for discontinuation (ie, 'AE', 'lack of efficacy', 'non-compliance' or 'other'); for these four trials, we will consider participants to have discontinued (dropped out) if their end-of-study status was not 'completed study/phase'.

Table 5 Variables used to calculate time under observation for each participant

Source	Variable name
CSR ^{22–24}	Last day in study
CSR ^{25 26}	Observation day
IPD	aestartday
IPD	aeendday
IPD	pain_d01 pain_d02 ... pain_d98
IPD ^{20–25 27}	sleep_d01 sleep_d02 ... sleep_d98
CSR ^{20 21}	Study day last double-blind medication
CSR ^{25 26}	Day of last dose of study drug

CSR, clinical study report; IPD, individual participant data.

- ▶ One trial²⁷ did not report end-of-study status in the CSR. Participants in this trial were assigned placebo for 1 week before random assignment to gabapentin or placebo for 14 weeks (98 days); thus, we will consider participants to have discontinued if their time under observation is less than the 98 days of treatment period.

Days with AEs

For each participant, we will calculate the number of days with AEs as the difference between the time of onset and time of resolution for each of the following levels: any AE, any serious AE, and specific AEs at level of preferred term, mid-level system and body system. Except for analyses at the level of preferred term, each day will be counted once. Thus, the number of days with AEs will not exceed the number of days of observation (eg, for each participant, a day with 'any AE' will be counted once whether the participant experienced one AE or multiple AEs on that day).

Consistency and data quality

To check the quality of the dataset, we will compare CSRs and IPD with regard to the time of onset of AEs, last medication time, time under observation and study period. We will flag any observations where the AE's time of onset is indicated to be at least 2 weeks (14 days) longer than the study period. These observations will be excluded from the primary analyses, but will be included in sensitivity analyses.

Risk of AE and risk of discontinuation

For the randomised participants, we will calculate the proportion (risk) of participants in each group reporting: any AE, any serious AE, each mid-level system, each body system and selected preferred terms. We will also calculate the proportion of randomised participants in each group who discontinued because of AEs and who discontinued for any reason. For these outcomes, we will report the differences between-groups as relative effects (eg, risk ratio (RR) or OR) and absolute effects (risk difference (RD)), including the corresponding 95% CIs.

Proportion of days with AE

For the randomised participants, we will calculate the proportion of days with any AE, any serious AE, selected preferred terms, each mid-level term and each body system. For these outcomes, we will report the ratio between groups.

The proportion of days with AE in each group is the total number of days with the AE divided by the total time under observation (eg, '5 days with headache per week'). Because randomised participants who do not take at least one dose of study medication will not contribute any person-time for this analysis, our planned analysis of the full intention-to-treat population will be identical to an analysis limited to the 'safety population'.

The proportion ratio is the proportion in the intervention group divided by the proportion in the placebo

group, which we will express (eg, 'two times more days with headache').

Incidence rate

For the randomised participants, we will calculate the incidence rate of any AE, any serious AE, selected preferred terms, each mid-level term and each body system. For these outcomes, we will report the incidence rate ratio (IRR) and 95% CI to compare the differences between two treatment arms. We will consider the same models proposed for 'risk of AE and risk of discontinuation'.

Time-to-event (survival analysis) for harms and discontinuation

For participants who took at least one dose of study medication (and whose time under observation is therefore greater than 0 days), we will use survival analyses to investigate differences in: any AE, any serious AE, each mid-level AE, each body system AE, discontinuation because of AEs and discontinuation for any reason, and selected preferred terms. For these outcomes, we will calculate the differences in median time-to-event, and we will report the HR and its 95% CI. Although we expect survival analyses to complement analyses based on risk, discontinuation is a competing risk for the reporting of AEs to trial investigators, so it is possible that RRs and HRs would differ in magnitude or direction. We will fit a stratified Cox regression model with random effects.⁴² The proportional hazard assumption will be tested using Grambsch and Therneau test and Schoenfeld residuals. If the assumption is not valid, we will consider alternative models such as reporting time-varying HR or cumulative Cox regression.⁴³

We will calculate time to event as the difference between randomisation and the day on which the AE started. For participants reporting the same AE more than once, we will use the time-to-first occurrence and we will exclude future occurrences from this analysis. Participants who discontinue the study will be censored from the time of study discontinuation. We will include the total time under the observation for participants who complete the study without reporting any AEs, or the AE of interest, as appropriate.

Statistical significance (p values and CIs)

For the effect estimates that we calculate, we will also calculate p values and CIs. Although p values and other methods of assessing statistical significance should not be used for null hypothesis testing with AEs, they can be helpful in aiding interpretation of results.^{44–47} While it is important to show some measure of uncertainty surrounding an estimate to provide a sense of the strength of the evidence for an association—especially as effect estimates for AEs can be high when events are rare—it is also important that they not be overinterpreted as proof or lack of proof of associations. Thus, we will not consider these to be null hypothesis tests and will not interpret 'significant' values as evidence of causal relationships.

Instead, we will interpret these values together with the effect estimates and number of events.

Many non-systematically assessed harms will be uncommon. Trials are rarely designed or powered to detect differences between groups in the occurrence of non-systematically assessed harms, unlike potential benefits. Uncommon events may produce unstable estimates wherein the bounds of the CI are unreasonable and hypothesis tests may be rendered invalid.⁴⁸ Moreover, hundreds of different non-systematically assessed harms might be reported, and statistical challenges are exacerbated by the problem of multiple testing, although some statistical methods have been developed to ameliorate these issues (eg, False Discovery Rate).^{48–50}

Methods for data synthesis

Combining effects across studies

We will estimate effects by combining evidence across studies using RRs, ORs, RDs and HRs and corresponding 95% CIs for: any AE, any serious AE, each mid-level AE, each body system, selected preferred terms, discontinuation because of AEs and discontinuation for any reason.^{42 51 52} We will conduct two-stage IPD meta-analyses. One-stage and two-stage meta-analyses will give almost identical results when models are not adjusted by baseline covariates, and we do not expect to adjust using baseline covariates because relatively little information is available from the included trials (see table 3).^{53 54} Specifically, the first stage will aggregate IPD for each trial and each arm. For the second stage, we will consider various meta-analytic methods to fit rare binary outcomes.⁵⁵ We will fit traditional frequentist methods including Peto and Mantel-Haenszel and Bayesian hierarchical meta-analytic methods that incorporate between-study heterogeneity with random effects. This method is preferable to a frequentist approach because studies with ‘zero’ cells are not a problem for the Peto, Mantel-Haenszel and Bayesian methods. For various approaches to handling such ‘zero’ cells for other frequentist meta-analysis models are proposed by Sweeting *et al.*⁵⁶ For the proportion of days with AEs, we will fit Bayesian Poisson regression with random effects to estimate the proportion ratio.

To identify a subset of preferred terms for reporting and for further analysis, we will conduct an interim analysis using the combined events in the gabapentin groups and the combined events in the placebo groups to calculate the frequency of each preferred term in the gabapentin group, and the RR, OR and the HR for the gabapentin group compared with the placebo group. We will focus on preferred terms associated with the gabapentin group (eg, RR >0 or HR >1.0). We will explore various methods for selecting AEs for further investigation and we will describe the implications of selection criteria for our results.

Level of analysis issues

For multiarm trials in which participants were randomised to different doses of gabapentin,^{21 25} we will combine the

gabapentin groups into a single group and compare it with the placebo group, as recommended in the Cochrane Handbook.⁵⁷

Assessment of heterogeneity

We will assess clinical and methodological heterogeneity using data coded by the MUDS investigators. We will interpret the results alongside the characteristics of studies, including risk of bias assessments, in tables and in a structured narrative.

To quantify statistical heterogeneity in results, we will calculate I^2 and perform Cochran's χ^2 test (ie, Q test), and we will visually examine the forest plots. An I^2 of over 75% will be considered as high heterogeneity, and we will adopt a p value of 0.10 as a threshold for statistical significance of Q test.⁵² We will also report estimated SD of random effects from Bayesian random effects models.

Subgroup analysis

Because women tend to be smaller than men, we expect that women might experience more AEs than men for any given dose of a drug. We will explore differences between sex (men compared with women) for discontinuation because of AEs and for each body system. Other exploratory subgroup analyses may be considered depending on the findings.

Sensitivity analysis

Because two trials^{21 25} randomised participants to multiple doses of gabapentin (eg, 600 mg/day, 2400 mg/day) or placebo, we will perform sensitivity analysis for selected outcomes by dose of gabapentin. For selected outcomes, we will ‘split’ randomly the shared placebo group into two or more groups with smaller sample size to account for the fact that the group has been used twice or more, and include each pairwise comparison separately in the meta-analyses across trials.⁵⁷ For example, in the three-arm trial of gabapentin 1800 mg, 2400 mg and placebo, we could compare 1800 mg gabapentin with half of the placebo group and 2400 mg gabapentin with the other half of the placebo group.

Analysing and reporting preferred terms

To identify a subset of preferred terms for reporting and for further analysis, we will conduct interim analyses in which we calculate the frequency of each preferred term in the gabapentin group and the treatment effects comparing the gabapentin group with the placebo group. We will focus on preferred terms that are associated with gabapentin (eg, RD >0.0, RR >1.0, IRR >1.0, HR >1.0). Of those, we will explore various methods for selecting AEs for further investigation and we will describe the implications of selection criteria for our results.

In each individual trial, we will attempt to calculate RRs, ORs, RDs, IRRs, and HRs for the selected preferred terms. If these results cannot be calculated in a given trial, we will report descriptive results (eg, no events were observed in any group). To synthesise the results across trials, we will calculate RRs, RDs and HRs for the preferred terms

selected for further analysis (see the ‘Combining effects across studies’ section).

STATISTICAL SOFTWARE

Analyses will be conducted, and figures will be drawn, in R statistical software (R V.4.0.3 and RStudio V.1.2.5001)⁵⁸ as needed.

ETHICS AND DISSEMINATION

We will use secondary data. This study was determined to be not human subjects research and thus exempt from IRB review (protocol #1910607198).

DISCUSSION

To make informed decisions about health interventions, patients and other stakeholders need accurate and complete information about both benefits and harms. Syntheses of clinical trial findings should include all available evidence; however, they are often only based on information reported in public sources, such as journal articles,^{59–61} which are often incomplete.^{5–7 61–72} Compared with journal articles, CSRs and IPD contain much more information about harms observed in clinical trials.^{73–75} Moreover, a comprehensive analysis of the harms associated with any commonly prescribed drug would include observational evidence, which might apply to relatively larger and more heterogeneous populations over longer periods of time.

The methods used to assess and to report harms in clinical trials contribute to challenges for interpreting and synthesising trial evidence, and many systematic reviews that plan to synthesise harms ultimately do not address them.⁷⁶ While AEs can be assessed systematically in clinical trials⁹—using methods like those used to assess potential benefits^{6 7}—AEs are often assessed non-systematically in response to open-ended questions such as ‘have you noticed any symptoms since your last visit?’ or by spontaneous reporting. Evidence syntheses (eg, systematic reviews, clinical practice guidelines) could help identify rare AEs if all observed AEs were available for all trials⁷⁷; however, rare AEs cannot be identified when clinical trials report only those AEs occurring above certain thresholds. Just as selectively reporting potential benefits based on quantitative results leads to biased meta-analyses^{78–81} selection criteria for reporting AEs will lead to biased overall estimates. At the same time, reporting hundreds of events might overwhelm patients and clinicians with information that does not help them make decisions about whether and how to use medicines.

This restoration will address both problems by publishing the complete AEs observed in these trials and producing a clinically informative summary following a prespecified Statistical Analysis Plan (SAP).^{1 7} Moreover, it will advance methods for analysing and reporting AEs in clinical trials. It is a limitation that this study will be

include only a subset of known gabapentin trials, and it is a limitation that the system used to categorise and analyse AEs in these trials (ie, COSTART) is no longer in regular use (ie, it has been replaced by MedDRA). We hope this project will facilitate future guidance for treating neuropathic pain, and it will help patients and clinicians make informed decisions about the use of gabapentin.

Author affiliations

¹Department of Epidemiology and Biostatistics, Indiana University School of Public Health–Bloomington, Bloomington, Indiana, USA

²Department of Epidemiology, Johns Hopkins University Bloomberg School of Public Health, Baltimore, Maryland, USA

³Department of Biostatistics and Bioinformatics, Duke University, Durham, North Carolina, USA

⁴Department of Biostatistics and Informatics, University of Colorado School of Public Health, Aurora, Colorado, USA

⁵Department of Ophthalmology, University of Colorado, Denver, Colorado, USA

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ORCID iD

Evan Mayo-Wilson <http://orcid.org/0000-0001-6126-2459>

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